

Unit 4 Practice MCQ Questions

- The equilibrium constant, K_c , for the reaction

$$\text{H}_2(g) + \text{I}_2(g) \rightarrow 2 \text{HI}(g) \quad \Delta H_{\text{rxn}} = -100 \text{ kJ/mol}$$

is 55 at 425°C. The value for the equilibrium constant will be increased if

 - the temperature is changed to 500°C.
 - a catalyst is added to the reaction system.
 - the size of a reaction vessel is doubled.
 - the temperature is changed to 400°C.
 - additional HI is added to the reaction system.
- Which of the following statements is true in a reaction system at equilibrium?

 - The number of collisions per unit time between reactants is equal to the number of collisions per unit time between products.
 - Reactants are reacting to form products at the same rate as products are reacting to form reactants.
 - The product of the concentrations of the reactants is equal to the product of the concentrations of the products.
 - All concentrations of reactants and products are the same.
 - No products are being formed, and no reactants are being formed.
- For the reaction system

$$\text{CoO}(s) + \text{H}_2(g) \rightarrow \text{Co}(s) + \text{H}_2\text{O}(g)$$

at 550°C, $K = 67$. The equilibrium constant expression is

 - $\frac{[\text{CoO}][\text{H}_2]}{[\text{Co}][\text{H}_2\text{O}]}$
 - $\frac{[\text{Co}][\text{H}_2\text{O}]}{[\text{CoO}][\text{H}_2]}$
 - $\frac{[\text{Co}][\text{H}_2\text{O}]}{[\text{H}_2]}$
 - $\frac{[\text{H}_2]}{[\text{H}_2\text{O}]}$
 - $\frac{[\text{H}_2\text{O}]}{[\text{H}_2]}$
- Carbon tetrachloride reacts with oxygen at high temperatures to produce chlorine and carbonyl chloride.²

$$\text{CCl}_4(g) + \text{O}_2(g) \rightarrow 2 \text{COCl}_2(g) + 2 \text{Cl}_2(g) \quad K_c = 1.9 \times 10^{19}$$

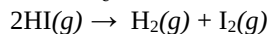
What is K_c for the following:

$$\text{COCl}_2(g) + \text{Cl}_2(g) \rightarrow \text{CCl}_4(g) + \frac{1}{2} \text{O}_2(g)$$
 - 5.2×10^{-20}
 - 2.3×10^{-10}
 - 4.4×10^9
 - -1.9×10^{19}
 - 1.9×10^{19}
- Consider the equilibrium $\text{N}_2(g) + 3 \text{H}_2(g) \rightarrow 2 \text{NH}_3(g)$ at a certain temperature. An equilibrium mixture in a 4.00-L vessel contains 0.40 M NH_3 , 0.200 M N_2 , and 0.30 M H_2 . What is the value of K_c ?

 - 9.00
 - 29.6
 - 3.37
 - 17.1
 - 7.41

6. For the reaction system:

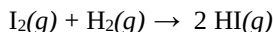
$$K_c = 0.020 \text{ at } 720 \text{ K.}$$



If the initial concentrations of HI, H₂, and I₂ are all $1.50 \times 10^{-3} \text{ M}$ at 720 K, which one of the following statements is correct?

- A. The system is at equilibrium.
- B. The concentrations of HI and I₂ will increase as the system is approaching equilibrium.
- C. The concentrations of H₂ and HI will decrease as the system is approaching equilibrium.
- D. The concentration HI will increase as the system is approaching equilibrium.
- E. The concentrations of H₂ and I₂ will increase as the system is approaching equilibrium.

7. In an experiment, 0.100 mol H₂ and 0.100 mol I₂ are mixed in a 1.00-L container and the reaction forms HI. If $K_c = 50.0$ for this reaction, what is the equilibrium concentration of HI?



- A. 0.156 M
 - B. 0.0780 M
 - C. 7.07 M
 - D. 0.200 M
 - E. 0.0384 M
8. Exactly 1.0 mol N₂O₄ is placed in an empty 1.0-L container and is allowed to reach equilibrium described by the equation
- $$\text{N}_2\text{O}_4(g) \rightarrow 2 \text{NO}_2(g)$$
- If at equilibrium the N₂O₄ has a concentration of 0.80 M, what is the value of the equilibrium constant for the reaction under these conditions?
- A. 0.05
 - B. 0.2
 - C. 0.5
 - D. 20
 - E. 400

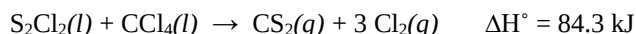
9. For the equilibrium



$K_c = 4.0$ at 228°C. Pure PCl₅ is added to the reaction system. At equilibrium there is 0.040 M PCl₅. What is the concentration of PCl₃ in this system?

- A. 0.16 M
 - B. 0.20 M
 - C. 0.40 M
 - D. 0.80 M
 - E. 1.6 M
10. Hydrogen iodide undergoes decomposition according to the equation
- $$2\text{HI}(g) \rightarrow \text{H}_2(g) + \text{I}_2(g)$$
- The equilibrium constant at 425°C for this system is 0.018. If 1.0 mol each of H₂, I₂, and HI were placed together in a 1-L container at 425°C, then
- A. the concentration of HI would decrease.
 - B. the value of K would increase to 1.0.
 - C. because of reaction, the total number of molecules would decrease.
 - D. because of reaction, the total number of molecules would increase.
 - E. the concentration of H₂ would decrease.
11. Which of the following equilibria would not be affected by pressure changes at constant temperature?
- A. $\text{FeO}(s) + \text{CO}(g) \rightarrow \text{Fe}(s) + \text{CO}_2(g)$
 - B. $\text{CaCO}_3(s) \rightarrow \text{CaO}(s) + \text{CO}_2(g)$
 - C. $\text{H}_2\text{O}(g) \rightarrow \text{H}_2(g) + \frac{1}{2} \text{O}_2(g)$
 - D. $2 \text{NO}(g) + \text{O}_2(g) \rightarrow 2 \text{NO}_2(g)$
 - E. $\text{PCl}_5(l) \rightarrow \text{PCl}_3(g) + \text{Cl}_2(g)$

12. Consider the reaction



If the above reactants and products are contained in a closed vessel and the reaction system is at equilibrium, the number of moles of CS_2 can be increased by

- A. adding some S_2Cl_2 to the system.
 - B. removing some CCl_4 from the system.
 - C. decreasing the size of the reaction vessel.
 - D. increasing the temperature of the reaction system.
 - E. adding some Cl_2 to the system.
13. The reaction of carbon monoxide and iodine pentoxide as represented by the equation
- $$5 \text{CO}(g) + \text{I}_2\text{O}_5(g) \rightarrow \text{I}_2(g) + 5 \text{CO}_2(g)$$
- is endothermic. The yield could be increased by
- A. increasing the pressure.
 - B. decreasing the pressure.
 - C. increasing the temperature.
 - D. decreasing the temperature.
 - E. decreasing the volume of the reaction vessel.
14. Which of the following equilibria would not be affected by volume changes at constant temperature?
- A. $2 \text{CO}(g) + \text{O}_2(g) \rightarrow 2 \text{CO}_2(g)$
 - B. $2 \text{NO}_2(g) \rightarrow \text{N}_2\text{O}_4(g)$
 - C. $2 \text{NO}(g) + 3 \text{F}_2(g) \rightarrow 2 \text{F}_3\text{NO}(g)$
 - D. $\text{BF}_3(g) + \text{NH}_3(g) \rightarrow \text{H}_3\text{NBF}_3(s)$
 - E. $\text{O}_3(g) + \text{NO}(g) \rightarrow \text{NO}_2(g) + \text{O}_2(g)$
15. For the reaction system $\text{H}_2(g) + \text{I}_2(g) \rightarrow 2 \text{HI}(g)$, $K = 55$ at 425°C .
For the system $\text{HI}(g) \rightarrow \frac{1}{2} \text{H}_2(g) + \frac{1}{2} \text{I}_2(g)$, K at 425°C is
- A. -55
 - B. 55
 - C. 0.13
 - D. 0.018
 - E. 3.3×10^{-4}
16. For which of the following systems at equilibrium and at constant temperature will decreasing the volume cause the equilibrium to shift to the right?
- A. $\text{N}_2(g) + 3 \text{H}_2(g) \rightarrow 2 \text{NH}_3(g)$
 - B. $2 \text{H}_2\text{O}(g) \rightarrow 2 \text{H}_2(g) + \text{O}_2(g)$
 - C. $2 \text{NO}_2(g) \rightarrow 2 \text{NO}(g) + \text{O}_2(g)$
 - D. $\text{NH}_4\text{Cl}(s) \rightarrow \text{NH}_3(g) + \text{HCl}(g)$
 - E. $\text{H}_2(g) + \text{Cl}_2(g) \rightarrow 2 \text{HCl}(g)$
17. In which of the following reactions does an instantaneous increase in the volume of the reaction vessel favor formation of the products?
- A. $\text{MgO}(s) + \text{CO}_2(g) \rightarrow \text{MgCO}_3(s)$
 - B. $\text{PCl}_5(g) \rightarrow \text{PCl}_3(g) + \text{Cl}_2(g)$
 - C. $\text{H}_2(g) + \text{I}_2(g) \rightarrow 2 \text{HI}(g)$
 - D. $\text{N}_2(g) + \text{O}_2(g) \rightarrow 2 \text{NO}(g)$
 - E. $\text{N}_2(g) + 3 \text{H}_2(g) \rightarrow 2 \text{NH}_3(g)$
18. Consider the following reaction: $2 \text{CO}(g) + \text{O}_2(g) \rightarrow 2 \text{CO}_2(g)$ $K_p = 2.9 \times 10^3$
If equimolar amounts of CO and O_2 gas are introduced into the system and the system is allowed to come to equilibrium, which statement is correct?
- A. $P_{\text{CO}} < P_{\text{O}_2} < P_{\text{CO}_2}$
 - B. $P_{\text{CO}} = P_{\text{O}_2} < P_{\text{CO}_2}$
 - C. $P_{\text{CO}_2} < P_{\text{CO}} < P_{\text{O}_2}$
 - D. $P_{\text{CO}_2} < P_{\text{CO}} = P_{\text{O}_2}$